

1 Geometry

Example 1.1 Pythagorean theorem

Find the length of the hypotenuse of a right triangle with sides 3 and 4.

Solution

Using the Pythagorean theorem,

$$c^2 = 3^2 + 4^2 = 25$$

therefore,

$$c = 5.$$

□

Example 1.2 Area of a circle

Find the area of a circle with radius r .

Solution

The formula for the area of a circle is

$$A = \pi r^2.$$

For $r = 3$,

$$A = 9\pi.$$

□

2 Algebra

Example 2.1 Solving a linear equation

Solve

$$2x + 5 = 13.$$

Solution

Subtract 5 from both sides:

$$2x = 8.$$

Divide by 2:

$$x = 4.$$

□

As shown in Example 1.1, the Pythagorean theorem can be used to find

unknown sides of right triangles.